

Paul Dutton

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Professional Summary

Recent graduate with machine learning research experience relating to Tiny ML on IoT devices, compressed sensing, and music generation. Diverse history with hardware, sales and analysis, and management of personnel. Looking to work in software engineering, data science, machine learning engineering, and GPU kernel development. Modular Champion.

Education

Bachelor of Science in Computer Science, The University of Texas at San Antonio Spring 2024 – May 2025
Data Science track. 4.0 GPA at graduation.
General Credit, Austin Community College Fall 2022 – Spring 2023
Bachelor of Science in Computer Engineering, Texas A&M, College Station Fall 2013 – Spring 2016

Research and Projects

- Implemented LeNet-5 (CNN) from scratch in Mojo for CPU and with custom, multi-streamed and fused GPU kernels. 91.4% throughput gains over PyTorch, 12.4% over ONNX-TensorRT (Nvidia) at same batch size on GPU. Custom logging, data loading pipelines. Presentation slated for ModCon 2026.
- TinyML On IoT. Convolutional Neural Network to detect heart VAs via ECG data on ARM Cortex-M4
 - Implemented data processing and model compression (SVD/QR pivoting, FFT, Wavelet Decomposition, Compressed Sensing). Personalized AI with per-patient patterning. Hyperparameter tuning, model modification. Maintained accuracy of reference model using just 10% of the original data points and model size; paper drafted
- Second project: generative AI (LLM based) for musical therapy applications utilizing biofeedback / EEG with CS, running on embedded devices. Custom tokenization for MIDI format.

Skills

Technical: Python, C, CUDA, Pandas, NumPy, Git, SQL, PyTorch, Bash, GPUs, Supercomputing, CNNs, SIMD.
Relevant Coursework: AI Hardware and Programming, High Performance Machine Learning, Cloud Computing, Data Science, Parallel Programming, Database Systems, Application Programming.

Work Experience

The Austin Shaker, Data Analytics, Wine Buyer, Manager. Austin, Texas September 2019 – December 2023

- Wine buyer for all locations. Managed a store, personnel, buying, led team, trained new managers
- Analyzed historical data with Pandas / Matplotlib; grew sales 83-100%, cleaned and fixed database (est. 21% of SKUs updated), automated intra-company transfers to fill gaps and move excess to prevent reorders
- Managed reorder levels using statistic data to save estimated \$180,000 across retail stores in on hand inventory

Tekmos, Test Engineer Intern. Austin, Texas May 2014 – February 2016

- Tested assembled packages for performance and operation, worked with predominately legacy hardware
- Tested hardware at high temperatures (300C +), wrote machine code to enhance stability tests, decreased false negatives by 15-20%, improving efficiency by reducing need to re-test as many packages

Siemens, Software and Data Intern for Green Energy (Site Controls). Austin, Texas May 2013 – August 2013

- Migrated database with Groovy / SQL. General data cleaning and validation
- Feature engineering, hashing IoT locations for optimal grouping, hardware fault detection, report generation
- Met with prospective client for sale of hardware infrastructure
- Web scraping competitors for price catalogs and comparisons with scripts, photographing items for our catalog